

Multiple Choice (10 marks)

1. An automobile travelling at 10 miles per hour produces 0.33 l b of CO gas per mile. How many moles of CO are produced per mile?
 - (a) 5.4 moles per mile
 - (b) 9.0 moles per mile
 - (c) 2.5 moles per mile
 - (d) 7.3 moles per mile
 - (e) 3.8 moles per mile
2. What weight of sulfur must combine with aluminum to form 600 lbs of aluminum sulfide?
 - (a) 350 lbs. sulfur
 - (b) 320 lbs. sulfur
 - (c) 425 lbs. sulfur
 - (d) 400 lbs. sulfur
 - (e) 300 lbs. sulfur
3. Sirius, one of the hottest known stars, has approximately a blackbody spectrum with $\lambda_{\text{max}} = 260 \text{ nm}$. Estimate the surface temperature of Sirius.
 - (a) 8500 K
 - (b) 15000 K
 - (c) 9500 K
 - (d) 11000 K
4. $\text{SO}_2 + \text{Cl}_2 \rightarrow \text{SO}_2 + \text{Cl}_2$ is 8.0 minutes. In what period of time would the concentration of $\text{SO}_2 + \text{Cl}_2$ be reduced to 1.0% of the original?
 - (a) 43 minutes
 - (b) 103 minutes
 - (c) 93 minutes
 - (d) 113 minutes
 - (e) 23 minutes
5. A gas of molecular weight 58 was found to contain 82.8% C and 17.2% H. What is the molecular formula of this compound?

- (a) C₃ H₆
- (b) C₅ H₁₀
- (c) C₄ H₈
- (d) C₄ H₉
- (e) C₃ H₈

True / False (10 marks)

Answer True or False

6. True or False: The final concentration of the mixed solution is 0.45 M H₂SO₄, indicating an increase in molarity due to dilution.
7. True or False: The number of configuration state functions in a full CI calculation of CH₂SiHF using a 6-31 G** basis set is 9.45×10^{27} .
8. True or False: Reacting 80 g of O₂ requires 72 g of NH₃, as it is a common misconception to overestimate the amount needed.
9. True or False: If the position of an electron is known within a 10 pm interval, the uncertainty in its momentum is $9.1 \times 10^{-23} \text{ kg m s}^{-1}$.
9. True or False: During the boiling of liquid NH₃, the hydrogen bonds holding separate NH₃ molecules together break apart.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the relationship between kinetic energy and wavelength for a photon and an electron, each possessing 1 eV of kinetic energy. Include calculations and implications of these wavelengths.
12. Discuss the weight of 1,000 cubic feet of air at Standard Temperature and Pressure (STP), including the calculations and implications of your findings in a broader scientific context.

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